

Industrial Robot

THE400-E Integration Manual

Rev. 2

November 30, 2022

Important

- Make sure that this instruction manual is delivered to the final user of Shibaura Machine's industrial robot.
- Be sure to read this manual and that you fully understand its contents before operating the industrial robot.
- After reading this manual, keep it in an easily accessible place for future reference.

Original Instructions

Shibaura Machine

This instruction manual has been prepared as "ANNEX VI Assembly instructions for partly completed machinery" under the Machinery Directive.

This manual is intended for the following robots.
THE Series: THE400-E

This manual is intended for the following controllers.
TS5000-EMS

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Introduction

Thank you for purchasing Shibaura Machine's robot system.
This manual describes how to integrate the Shibaura Machine System Robot THE400-E into equipment.
To fully understand the performance of the robot and to use its functions safely for a long time, please read this manual carefully, and operate the robot by following the instructions.

This manual has the following chapters.

- Scope of Use
This explains the intended scope of use for this product.

- Usage Procedures
This explains the main specifications of this product and how to use it safely.

- Warranty
This explains the terms and conditions of the warranty.

- Standards, Laws, and Regulations Pertaining to the Safety of Industrial Robots
This explains the main standards, laws, and regulations that must be followed when using the robot.

- Service Network
This provides the contact information for technical support and requesting service.

- Integration Procedure
This section describes the procedure for safely integrating the THE400-E into your company’s equipment.

- Related Standards
Typical related standards that we recommend referencing when integrating the THE400-E.

Please have the following instruction manual ready when reading this manual.

Document Number	Title
STEA1213	STEA1213 THE400/TS5000 Series Robot Manual
STEA1223	STEA1223 TS5000-EMS Controller Manual
STEA1214	THE400-E Safety Manual
STEA1225	STEA1225 TS5000-EMS Interface Manual
STEA1162	STEA1162 TP5000 Operation Manual
STEA1163	TS5000 Robot Language Manual (SCOL2)
STEA1165	TS5000 Alarm Manual

Manufacturer

Shibaura Machine Co., Ltd.
2068-3, Ooka, Numazu-shi, Shizuoka-ken 410-8510, Japan
Tel: 81-(0)55-926-5032
Fax: 81-(0)55-925-6527

If you have purchased this robot outside of Japan, please contact the company that you purchased it from.

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Scope of Use

This product is a horizontal articulated industrial robot, and the mechanical interface at the tip of the manipulator having two joints that turn in the horizontal direction moves up and down in the vertical direction and rotates in the horizontal direction.
The product has been developed for use in a wide range of automation tasks such as transport work including the movement and alignment of workpieces and assembly work such as insertion of parts and screw tightening.
The customer should also use this product for the purpose of automating these tasks and for incorporating into automation equipment.

This product is neither intended for use in tasks involving human contact (such as tasks for medical care and nursing care), nor for integration into equipment that sustains human life (such as life support devices and incubators), equipment on which people ride (including transportation equipment such as cars and aerospace equipment such as airplanes, and recreational equipment such as roller coasters), and equipment that protects human life or the human body (such as safety devices).
This product is not intended to be incorporated in equipment for controlling large heat quantities (such as nuclear power controllers and incinerator controllers).
This product is prohibited from being used for tasks or incorporated in equipment where a failure or malfunction could likely pose a direct threat to human life or harm to the human body. The customer is responsible for providing any warranties if this product is used.

Usage Procedures

This section describes the usage requirements for the THE Series. Usage of the product in a state outside the required conditions could result in unexpected risks. Be sure to always use the robot in an environment that meets the requirements

1. Main Specifications of Product: Refer to "1.4 Specifications Table" in the Robot Manual: SPECIFICATIONS and "3.1 Installation Environment" in the Robot Manual: INSTALLATION.
2. Usage Restrictions: Tables 1 to 7 provide the “Restricted Specifications Sheet” indicating the usage restrictions.

Table 1. Restricted Specifications Sheet

1. Purpose and application for using product (usage restrictions)	
1) Intended use	To make the robot perform operations for specific purposes (item transport, assembly, inspection, machining, etc.) (1) Transportation [1] Obtain a transportation route to the unpacking site. [2] Use a forklift or other equipment to lift the packing box. [3] Move the packing box to the unpacking site. (2) Unpacking [1] Obtain the space required for unpacking. [2] Cut the packing straps and unpack. [3] Cut the packing straps securing the robot. [4] Peel off the tape from the controller packing box, and unpack. [5] Check that the robot and controller are not damaged. [6] Check the accessories and their quantities. - When transferring the robot to the installation site [7] Dispose of, or store and manage the packing materials. - When storing the robot [7] Secure in place in the same way as the state at delivery, and repack. [8] Move the packing box to the storage site. (3) Transfer - Robot [1] Check the orientation of the robot during transfer. [2] Check that the robot fixing jigs are mounted. [3] If using a crane, mount the supplied eyebolts. [4] Check the tightness of the robot fixing jig fixing bolts. [5] Use a crane or cart to transfer to the installation site. - Controller [1] Use a cart to transfer to the installation site.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
1) Intended use (continued)	(4) Installation [1] Use bolts to secure the robot to the frame and other parts. [2] Fabricate any needed cables. [3] Remove the robot fixing jigs and eyebolts. [4] Install the controller and connect it to the robot. [5] Push the robot by hand to move to a location where it is easy to perform work. [6] Mount the hand. [7] Connect the required external devices. (5) Teaching [1] Move the robot to the desired position or orientation by guiding it with the teaching pendant or pushing it by hand. [2] Teach the position where the robot will operate. [3] Create the operation sequence (program, etc.). [4] Make sure that the signals and data are transferred to and from the external equipment. [5] Check operation at low speed (250 mm/sec or less). [6] While gradually increasing the operating speed, make sure that the operation matches the specific purpose. (6) Operation [1] Start robot operation. [2] Make sure that operation is being performed to match the specific purpose at the specified speed. [3] Stop robot operation. (7) Repacking [1] Remove the hand. [2] Move the robot to the position and orientation where the fixing jig can be mounted by guiding it with the teaching pendant or pushing it by hand. [3] Mount the fixing jig. [4] Turn off the power, and remove the wiring. [5] Obtain the space needed for packing. [6] Use a crane to transfer the robot, and mount the supplied eyebolts. [7] Transfer the robot and controller to the packing site. [8] Pack the robot and controller so that they are in the original delivery state. [9] Obtain a transportation route to the transportation equipment (truck, etc.). [10] Use a forklift to lift the packing box. [11] Load the packing box onto the transportation equipment.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
1) Intended use (continued)	(8) Maintenance Work performed when the primary power is off: [1] Check the bolts for any looseness and backlash and re-tighten them. [2] Make sure that the cable clamp is not loose. [3] Check for any damage. [4] Clean off any dust. [5] Inspect and apply corrosion inhibitor. [6] Check for any wear on the cable and air tube. [7] Check that there is no misalignment between the motion of each joint axis and the origin position. [8] Replace the memory backup batteries in the controller. Work performed when the primary power is on: [1] Check the grease state and apply grease (when the servo power is off). [2] Check for any abnormal vibrations and sounds during operation (when the servo power is on). [3] Check the teaching point for any misalignment (when the servo power is on). [4] Replace the backup batteries for the robot position detector (when the servo power is off). (9) Disposal [1] Remove the hand. [2] Move the robot to the position and orientation where the fixing jig can be mounted by guiding it with a teaching pendant or pushing it by hand. [3] Mount the fixing jig. [4] Turn off the power, and remove the wiring. [5] Use a crane to transfer the robot, and mount the supplied eyebolts. [6] Transfer the robot and controller to the disposal site. [7] Dispose of the robot using a method compliant with your country’s laws and regulations.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
2) Foreseeable misuse	- Use for work which applies excessive impact to the robot or to which excessive impact is applied by the robot - Modification of robot, controller, and optional parts, or supplying power to robot, controller, and optional parts while their robot covers are removed - Operation in other than the specified installation state (floor, suspended, and wall-mounted) - Operation on a frame or other structure that does not satisfy the specified strength - Operation under conditions not satisfying the specified installation requirements - Entry into the hazard zone of the robot by mistakenly assuming that operations would have stopped although the stop is just for waiting for an operation signal during operation - Usage under conditions where the specification values are exceeded or usage under improper settings - Entry into the hazard zone of the robot when the servo power is on - Stop of the robot caused by power off during operation or when the servo power is on - Unintended operation caused by inadvertently pressing of a button - Operating without checking operation in MANUAL mode - Operating without checking the tightness of the robot and hand set bolts - Unintended operation due to improper wiring - Unintended operation due to incorrect program selection - Riding on or leaning against the robot or hanging on the arm
3) Unexpected start	- Malfunction due to excessive noise - Robot start due to start signal unexpectedly sent from peripheral equipment to the robot - Unintended start due to abnormal communication data - Malfunction due to voltage fluctuation - Power on due to powering after power outage - Malfunction due to failure or other cause

Table 2. Restricted Specifications Sheet

2. Replacement or other modification of program components (time restrictions)	
1) Mechanical restrictions	- Before operating the robot, perform operation at a low speed (20% or less of maximum speed) for five (5) minutes - Replace consumable parts as required or periodically by conducting daily inspection, regular inspection (every 3 and 6 months and every year) and overhauls (every 5 years).
2) Electrical restrictions	

Table 3. Restricted Specifications Sheet

3 Movable range of product (restriction on space)		
1) Movable range	1st axis 2nd axis 3rd axis 4th axis Additional axis - -	Compliant with 1.4 Specifications Table Compliant with 1.4 Specifications Table Compliant with 1.4 Specifications Table Compliant with 1.4 Specifications Table Compliant with individual specifications. - -
2) Interface	Transportation: The packing box shall be able to be transferred by a transportation worker. Unpacking: The packing box shall be able to be carried in. Space of at least 2 m3 shall be secured. Transfer: The robot shall be able to be hoisted by a crane. The robot shall be able to be lifted by two or more people and loaded onto a cart or similar device. There shall be no obstacles on the route from the unpacking site to the installation site. Installation: The product shall be installed at a location suitable for the outline and work envelope of the robot. The product shall be installed at a location suitable for the outline and that provides adequate ventilation of the controller. The cable shall be installed at a location where it will not be stepped on by operators. Teaching: Teaching work shall be able to be performed from outside the hazard zone of the robot. Operation: A fence or enclosure shall be installed to prevent any operator from approaching the hazard zone of the robot. The robot shall be able to be operated from outside the hazard zone of the robot. An operator in the fence or enclosure shall be visible from the operation location. Repacking: Work shall be able to be performed to remove the hand, mount the fixing jig, separate the robot and controller, and remove the robot and controller from the frame. The robot shall be able to be moved from the installation site to the packing site. The repacking work shall be able to be performed. A forklift or other equipment shall be able to be used to lift the packing box. Maintenance: Space required for maintenance shall be obtained. After maintenance, operation shall be able to be checked from outside the hazard zone of the robot. Disposal: Work shall be able to be performed to remove the hand, mount the fixing jig, separate the robot and controller, and remove the robot and controller from the frame. The robot shall be able to be moved from the installation site to the disposal location.	

Table 3. Restricted Specifications Sheet

3. Movable range of product (restriction on space)	
3) Work environment	- A fence, enclosure, or other protection shall be provided that surrounds the hazard zone of the robot to prevent people from entering the zone during operation. - A display indicating the robot operation status shall be provided to indicate whether the robot is operating or stopped. - The driving power supply shall be turned off whenever a person enters the hazard zone of the robot. - Lighting shall be provided for enabling visual checks by operators. - To prevent operators from slipping, the floor shall not be covered with oil or similar substances. - No noises that impair attention shall be emitted.

Table 4. Restricted Specifications Sheet

4. Life cycle of product
In the product life cycle of assembly, shipment testing, transportation, unpacking, transfer, installation, teaching, operation, repacking, maintenance, and disposal, risk assessment is carried out for the life cycle of “transfer and installation” (transportation, unpacking, transfer, installation, and repacking), “use” (teaching, operation, and maintenance), and “disposal.”

Table 5. Restricted Specifications Sheet

5. Person who approaches the machine		
Applicable person	Knowledge and requirements	
1) Transportation worker	Knowledge	Contents of the instruction manual
	Requirement	Person who has received training for forklifts and other transfer equipment work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
2) Unpacking worker	Knowledge	Contents of the instruction manual
	Requirement	—
3) Transfer worker	Knowledge	Contents of the instruction manual
	Requirement	Person who has received training for crane work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
4) Installation worker	Knowledge	Contents of the instruction manual, operating procedures for target robot
	Requirement	When guiding a robot, a person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
5) Teaching worker Work supervisor	Knowledge	Contents of the instruction manual, operation procedures for target robot, work regulations, and robot operations
	Requirement	Person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
6) Operator	Knowledge	Contents of the instruction manual, operating procedures for target robot, work regulations, and robot operations
	Requirement	—
7) Repacking worker	Knowledge	Contents of the instruction manual, operating procedures for target robot
	Requirement	Person who has received training for crane work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer When guiding a robot, a person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer

Table 5. Restricted Specifications Sheet (continued)

5. Person who approaches the machine		
Applicable person	Knowledge and requirements	
8) Maintenance worker Work supervisor	Knowledge	Contents of the instruction manual, operating procedures for target robot, work regulations, and robot operations Storage procedures for target robot
	Requirement	Person who has received training for robot testing work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
9) Disposal worker	Knowledge	Contents of instruction manual, operating procedures for target robot, and waste disposal methods
	Requirement	Person who has received training for crane work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer When guiding a robot, a person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
10) People passing nearby Other workers	- Measures shall be taken to prevent entry into the fence, enclosure, or other protection surrounding the hazard zone of the robot. - Measures shall be taken to make the robot stop operation when someone approaches the hazard zone. - If another worker enters the hazard zone of the robot to perform work, robot operation shall be stopped, the driving power supply turned off, and a teaching worker or maintenance worker shall supervise the situation to ensure safety.	

Table 6. Restricted Specifications Sheet

6. Other

Warranty

We conduct strict testing and inspection of this product before delivery to ensure that it satisfies our performance standards.
We warrant this product to be free from defects under the following terms and conditions.

1. Warranty period

The warranty period of the product is one of the following, whichever is shorter.

1) 18 months after shipping from our factory
2) 12 months after installation at customer site

2. Warranty scope

1) The warranty is provided for this product. The warranty covers only the specifications and functions included in the product specifications, catalog, and instruction manual. Any secondary or incidental damages caused by failure of this product are not covered by the warranty under any circumstances.
2) We will repair a faulty product free of charge during the warranty period only if this product has been operated and used in accordance with the instruction manual supplied with the product.
   A fee will be charged for any repairs after the warranty period has expired.

3. Disclaimers The warranty does not include the following cases.

a) Any failure or damage arising from careless use or misuse, or usage that is not described in the instruction manual.
b) Any failure caused by aging or normal wear and tear (discoloration, deterioration of consumables[*1]).
c) Irregularities confined to sensory quality such as noise not recognized as affecting functions
d) When any modification or disassembly without our consent was performed
e) Any failure or damage arising from inadequate maintenance and inspection or improper repair
f) Any failure or damage due to an act of God, fire, and other force majeure
g) Internal data such as programs and points created or modified by customers
h) When a product purchased in Japan is taken abroad

4. Safety Information

1) Shibaura Machine shall provide no guarantees for the standard performance of this product when this product is used outside the specifications.
2) If the customer does not observe the safety information appearing for DANGER, WARNING, and CAUTION contained in this instruction manual, Shibaura Machine

will not be liable for any resulting injuries, death, or other accidents, damage, product failure, or any other problems.

3) The safety information appearing for DANGER, WARNING, CAUTION and other explanations in this instruction manual are within the scope of usage assumed by Shibaura Machine, and they do not foresee all possible incidents for all situations that could occur. Please be aware that the scope of this safety information is limited.

*1): Consumable parts refer to the service replacement parts as listed in the instruction manual "THE400-E Robot Manual: MAINTENANCE".

Standards, Laws, and Regulations Pertaining to the Safety of Industrial Robots

The international industrial standard ISO/DIS12100 "Safety of machinery" specifies risk reduction measures for machinery and equipment.
Step 1: Inherently safe design measures - Limitations on force, speed, energy, etc.
Step 2: Safeguarding - Installation of safety fences, etc.
Complementary protective measures - Installation of emergency stop devices, etc.
Step 3: Information for use - Warning signs, alarms, instruction manuals, etc.

Based on the above, the ISO/IEC has defined "Guide 51", which classifies the various safety standards in a hierarchal structure. Safety standards for industrial robots are standardized as separate safety standards for machinery in Type C.
●ISO 10218, ISO 10218-1

The providing of residual risk information to users after risk assessment has been conducted is standardized.
●IEC 82079-1

Be sure to always follow the standards, laws, and regulations set forth in the country where the industrial robot is used.

Safety Information

This instruction manual contains important information to ensure safe and correct use and to prevent injury to the operator and others and damage to property.
In order to ensure safety from robot installation to disposal, be sure to carefully read this manual before starting the actual work.
Be sure that you fully understand the following information (signal words and symbols) before reading the main text, and always follow the provided safety instructions.

Display of Signal Words and Their Meanings

Signal word and symbol	Meaning
⚠ DANGER	Indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury.
⚠ WARNING	Indicates a potentially hazardous situation which, if not avoided, will result in death or serious injury.
⚠ CAUTION	Indicates the possibility that incorrect operation could cause injury[1] to operator.
NOTICE	Indicates the possibility that incorrect operation could cause property damage[2] and advice about performing work.

1) Injury refers to wounds, burns, electric shocks, and similar cases that do not require hospitalization or prolonged outpatient treatment.
2) Damage to property refers to extensive damage to property or materials.

Scope of Integration Work

The expected work scope for each life cycle is shown below.
In actuality, some tasks cannot be clearly classified and others are spread over multiple life cycles. These classifications are intended as a general guideline only, and so please check the safety information for all life cycles , and take protective measures before starting work.

- Transportation: From unloading packing boxes brought by transportation equipment (truck, etc.) to carrying them to the unpacking location.

- Unpacking: From opening the packing box to taking out the product.

- Transfer: From taking out the product from the packing box to carrying it to the installation site.

- Installation: From installation of the product, wiring, piping, assembly of the hand and optional parts until it is powered on.

- Teaching: Changing parameters, teaching, programming, and checking operation.

- Operating: Up to operation check of equipment into which the robot has been integrated

Service Network

Please refer to our distributor's handouts

1. Integration Procedure

Robots are used by integrating them into automation equipment for a wide range of tasks, such as transport work such as moving and aligning workpieces, and assembly work such as inserting parts and tightening screws.

This section explains the exact method of integrating robots into equipment and precautions so as not to endanger the operators.

At the same time, please refer to the following manual.

- STEA1214 THE400-E/TS5000-E Safety Manual, 2.3 Examples of Safety Measures
  3. Troubleshooting

1.1 Transportation

When transporting the robot, please refer to the following manuals.

- STEA1214 THE400-E/TS5000-E Safety Manual, 3. Transportation
- STEA1223 TS5000-EMS Controller Manual, 2.1 Transporting the Controller

1.2 Unpacking

When unpacking the robot, please refer to the following manuals.

-STEA1213 THE400/TS5000 Series Robot Manual, 5. Unpacking

- STEA1223 TS5000-EMS Controller Manual, 2.2 Unpacking (Repacking) the Controller

1.3 Transfer

When transferring the robot, please refer to the following manuals.

- STEA1213 THE400/TS5000 Series Robot Manual, 4. Transfer
- STEA1223 TS5000-EMS Controller Manual, 2.2 Transferring the Controller

1.4 Installation

When installing the robot, please refer to the following manuals.

- STEA1213 THE400/TS5000 Series Robot Manual, 6. Installing the Robot
- STEA1223 TS5000-EMS Controller Manual, 3. Installation

The following explanations are based on an example of integrated equipment.

1.4.1 Installing the Robot onto the Frame

When assembling the robot onto the frame, please refer to the following manual.

-STEA1213 THE400/TS5000 Series Robot Manual, 6.2 Installation

1.4.1.1 Preparation of the Frame

Prepare a stand on which the robot will be installed that meets the following specifications.

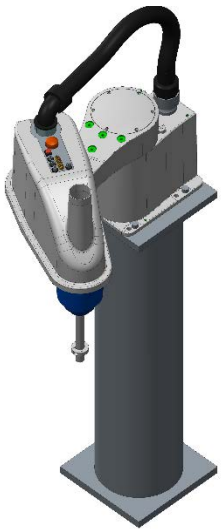
Load on the frame during horizontal movement (Nm)	223
Flatness of the robot installation surface	0.02
Surface roughness of the robot installation surface	Ra 6.3

1.4.1.2 Securing the Robot

Using the fixing holes on the base of the robot, secure the robot onto the frame.

1.4.1.3 Removing the Transportation Jigs

After securing the robot, remove the jigs used during transportation. Please keep the jigs as they will be used for repacking and disposal of the robot.



1.4.2 Integrating the Controller to the Control Panel

When assembling the controller to the control panel, please refer to the following manual.

- STEA1223 TS5000-EMS Controller Manual, 3. Installation

1.4.2.1 Securing Space in the Control Panel

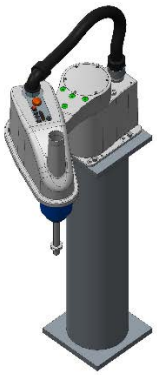
Prepare the control panel that can secure ventilation space for the controller so that heat does not accumulate.

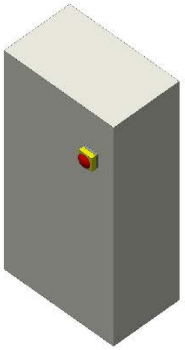
Be careful not to apply external force to the cable connected to the front of the controller when the door is closed.

1.4.2.2 Installing the Control Panel

Install the control panel in which the controller is housed outside the safety protection space, and set the operation location so that you can see the inside of the safety protection space.

Regarding safety protection spaces, see 1.4.5 Installing Guards





1.4.3 Connecting between the Robot and the Controller

Connect the robot and the controller.

1.4.3.1 Checking the Combination

There are designated combinations of robots and controllers for each model. Please refer to the following manual to make sure that the combination is correct.

- STEA1214 THE400-E/TS5000-E Safety Manual, 1. Warning Labels

1.4.3.2 Creating an 200 VAC Power Cable

Create a power cable that supplies 200 VAC power to the controller.

Please refer to the following manuals when performing the work.

- STEA1223 TS5000-EMS Controller Manual, 4.1.2 Connecting the AC Power Cable “AC IN”, Figure 4.1-①
- STEA1225 TS5000-EMS Interface Manual, 2.1 Connecting the Power Cable

1.4.3.3 Creating a 24 VDC External Power Cable for Input / Output Signals

Create a power cable that supplies 24 VDC for I/O signals to the controller.

Please refer to the following manuals when performing the work.

- STEA1223 TS5000-EMS Controller Manual
  4.1.4 Connecting the Input / Output External Power Supply and the Safety Input / Output Connector “SAFE”, Figure 4.1-⑥

1.4.3.4 Connecting Robot Control Lines, etc.

Connect the robot or teaching pendant to the controller.

When connecting, please refer to the following manuals.

- STEA1223 TS5000-EMS Controller Manual, 4.1.3 Robot Control Cable Figure 4.1-② to ⑤
- STEA1225 TS5000-EMS Interface Manual 3. Robot Control Cable
- STEA1162 TP5000 Operation Manual 2. Starting TP5000

1.4.3.5 Grounding

Ground the robot and controller.

When grounding, please refer to the following manuals.

- STEA1213 THE400/TS5000 Series Robot Manual, 6.2.2 Grounding Procedure
- STEA1225 TS5000-EMS Interface Manual, 1.11 Ground Terminal, Figure 1.1-⑭

1.4.3.6 Changing to MANUAL Mode

Toggle the master mode key switch on the controller to set the MANUAL mode.

When setting, please refer to the following manual.

- STEA1162 TP5000 Operation Manual, 3. Operation in MANUAL mode

1.4.3.7 Checking Normal Startup

After connecting the power cable created in 1.4.3.2 and 1.4.3.3 to the controller, turn on the power and check that it starts normally.

When starting, please refer to the following manual.

- STEA1223 TS5000-EMS Controller Manual, 5.1 Controller Operation Check

1.4.3.8 Checking and Canceling the Alarm

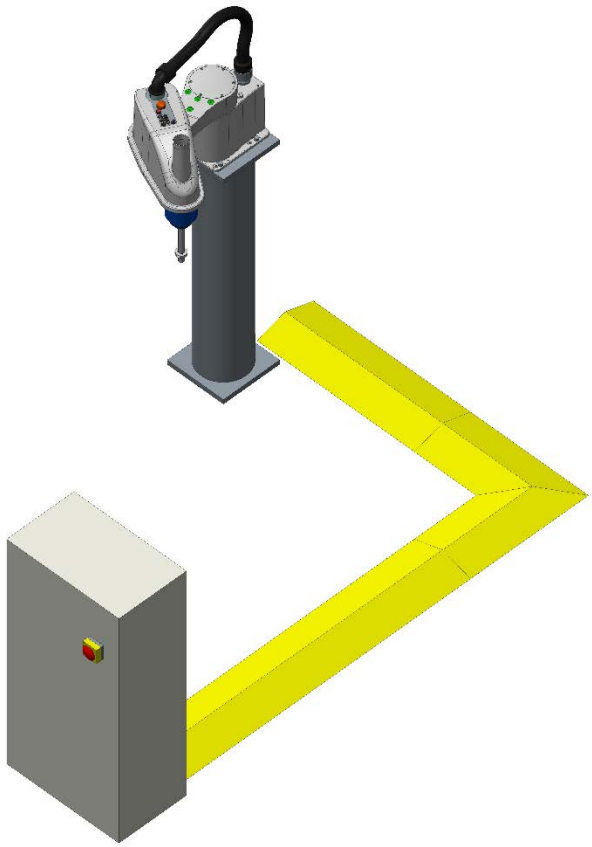
Check the contents of the alarm generated at startup on the TP5000. After removing the cause, cancel the alarm. It is also possible to stop only the buzzer sound of the alarm for which the cause cannot be removed.

When checking and canceling the alarm, please refer to the following manuals.

- STEA1162 TP5000 Operation Manual, 7. Operation in the Alarm Window
- STEA1165 TS5000 Alarm Manual

1.4.3.9 Protecting Cables

Leaving the cables on the floor can cause the operator to trip or fall. Protect each cable by putting it in the pit or covering it with a molding.



1.4.4 Restricting the Work Envelope

The robot has a mechanical stopper to limit the work envelop, which can be changed. If an unused range appears in the robot's work envelop due to integration into equipment, attach a newly created mechanical stopper to limit the work envelop to reduce the risk.

1.4.4.1 Restricting the Work Envelope

When limiting the work envelop with a mechanical stopper, please refer to the following manual.

- STEA1213 THE400/TS5000 Series Robot Manual, 6.3.1 Changing the 1st/2nd Axis Work Envelop
  6.3. 2 Changing the 3rd/4th Axis Work Envelope

1.4.4.2 Changing the Software Limit

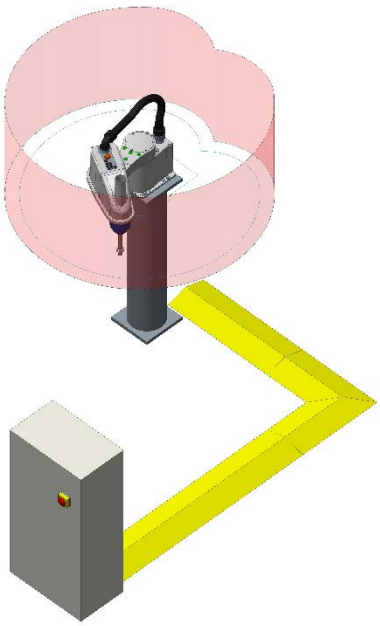
Change the software limit to prevent the robot from colliding with the mechanical stopper while the robot is being operated. The software limit cannot be used as a safety function to prevent collision with people.
When changing the software limit, please refer to the following manual.

- STEA1213 THE400/TS5000 Series Robot Manual, 6.3.3 Changing the Software Limit

1.4.4.3 Checking the Work Envelope

The 1st, 2nd, and 4th axes of THE400-E can be moved by turning off the driving power supply (turning off the servo) and pushing them by hand. By releasing the brake while the 3rd axis is powered on, it can be pushed and moved by hand without turning on the driving power supply (turning on the servo).
If you change the mechanical stopper, move the robot and check whether the work envelop is correctly limited and the software limit is changed.
When moving, please refer to the following manual.

- STEA1213 THE400/TS5000 Series Robot Manual, 9.1 Guiding the Robot by Pushing It by Hand



1.4.5 Installing the Guard

The space in which the work envelope of the robot is surrounded by a fence or the like to prevent people from approaching the robot containing the driving power supply is called a safety protection space. Install it before turning on the driving power supply of the robot (turning on the servo).

1.4.5.1 Installing the fence

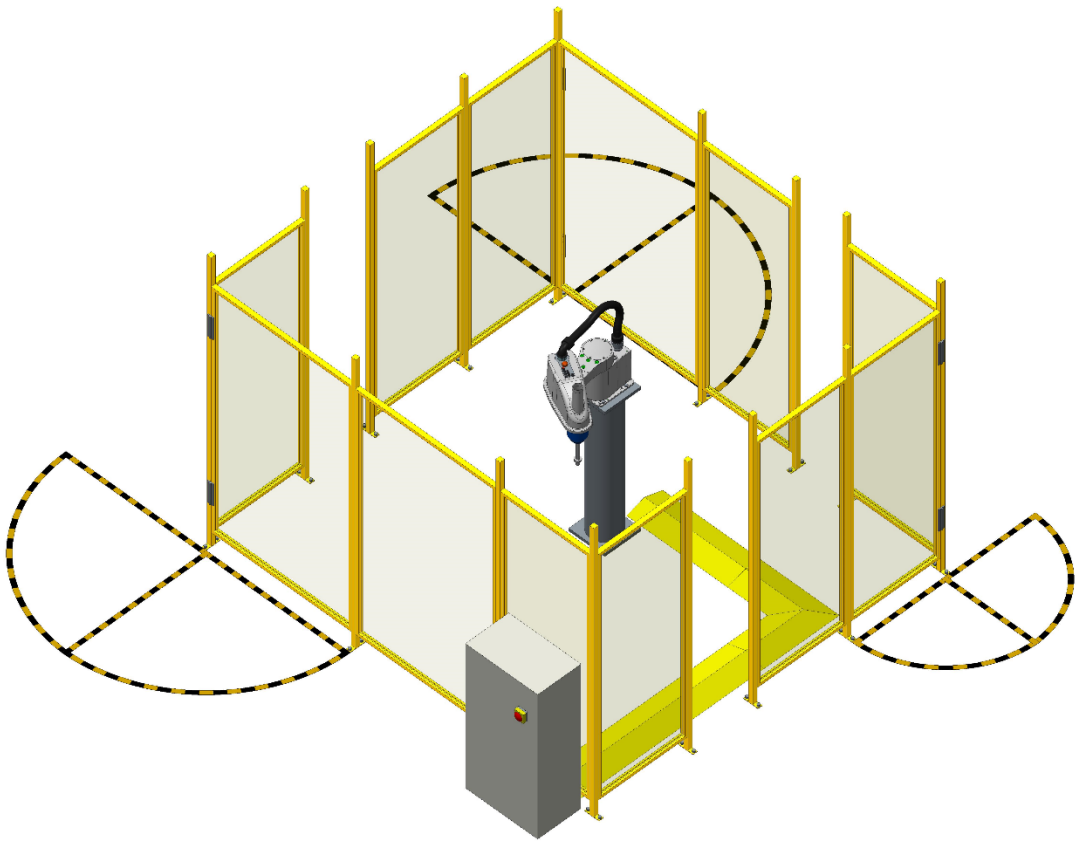
This fence is also called a fixed guard. Make sure there is enough space inside the enclosure for work before installing the fence.

The space to be secured varies depending on the equipment in which the robot is integrated. Please refer to the following manuals.

- STEA1213 THE400/TS5000 Series Robot Manual, 12.3 Preparation for Maintenance Work
- STEA1214 THE400-E/TS5000-E Safety Manual, 2.2.2 Robot Behavior during Emergency Stop

1.4.5.2 Installing the Safety Door

These door is also called a movable guard. Prepare a door with a safety switch (safety door) that complies with the standard, and install it at the entrance/exit of the safety protection space. When a person enters the safety protection space through the safety door, the robot is stopped and the servo is turned off. Connect the safety input signal from the safety relay unit to the safety door. Turn off the power before connecting the wiring. When connecting the wiring, please refer to 1.4.6 Connecting Safety I/O signals.

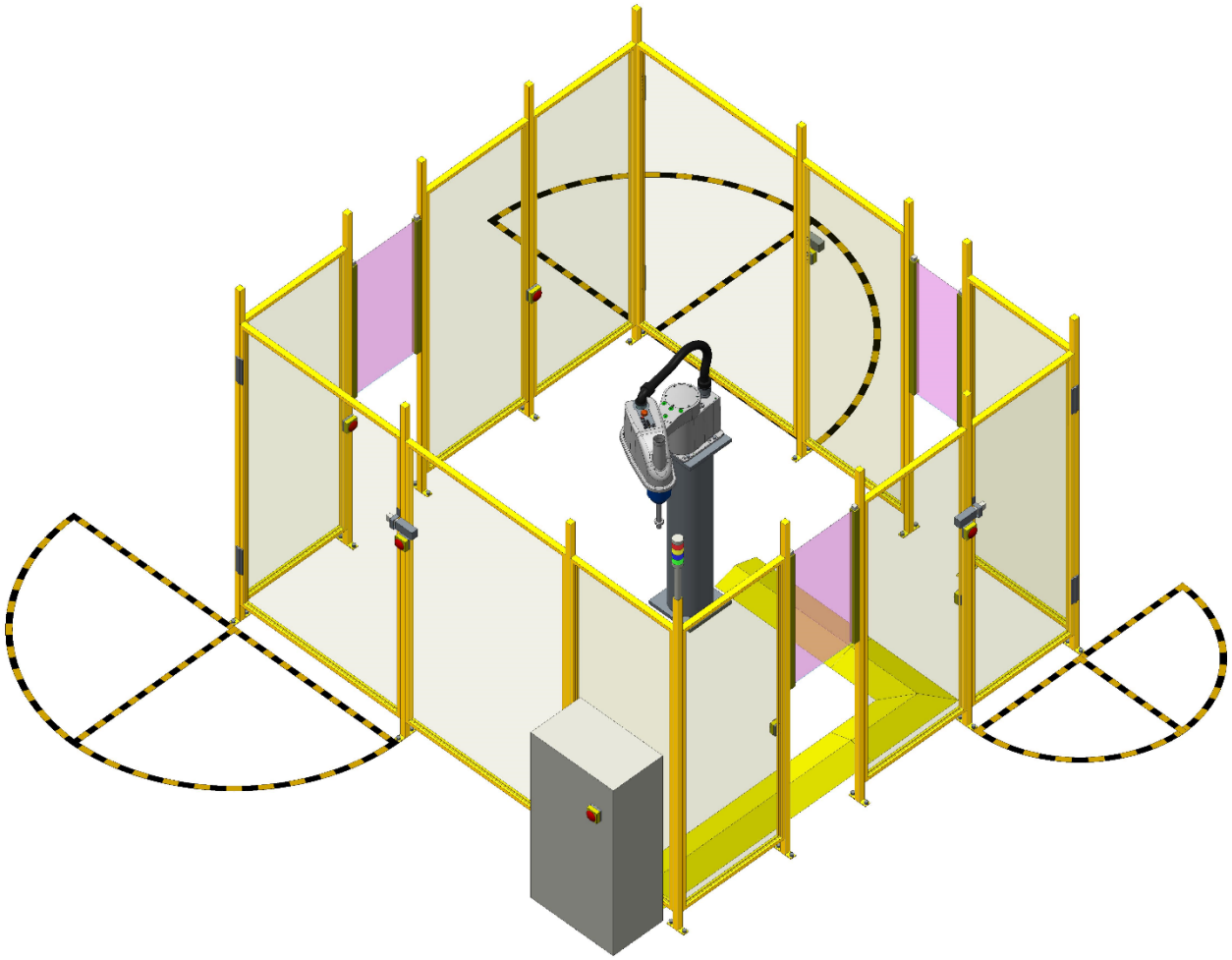


1.4.6 Connecting Safety I/O signals

Connect the robot to safety protective equipment such as a safety door, emergency stop button, and safety light curtain. Please refer to the following manuals when performing the work. Turn off the power before connecting the wiring.

Connect the wiring before turning on the driving power supply of the robot (turning on the servo).

- STEA1223 TS5000-EMS Controller Manual
  4.1.4 Connecting the Input / Output External Power Supply Cable and the Safety Input / Output Connector “SAFE”, Figure 4.1-⑥
- STEA1225 TS5000-EMS Interface Manual, 4. Input / Output External Power Supply Cable and Safety Input / Output Cable



1.4.7 Connecting with Peripheral Equipment

Connect the equipment to be used with the robot. Prepare peripheral equipment that complies with the respective safety standard.

1.4.7.1 Installing and Connecting the End Effector, etc.

The end effector is also called a robot hand or hand. Prepare the end effector according to its usage and attach it to the tool flange of the robot. A camera and air valve, for example, can be attached to the second arm of the robot.

Refer to the following manual when installing the end effector, creating cables, wiring, piping, and setting. Turn off the power before connecting the wiring to the robot.

- STEA1213 THE400/TS5000 Series Robot Manual,7. Hand Settings.
  8. User Wiring

1.4.7.2 Connecting with Peripheral Equipment

Connect the equipment used with the robot.

When wiring, please refer to the following manuals. Turn off the power before connecting the wiring to the robot and controller.

- STEA1213 THE400/TS5000 Series Robot Manual, 8.1.2 User Wiring
- STEA1225 TS5000-EMS Interface Manual, 1.5 Input Signal Cable “INPUT”, Figure 1.1-⑧
  1.6 Output Signal Cable “OUTPUT”, Figure 1.1-⑨
  5. Input Signal Cable
  6. Output Signal Cable

1.4.7.3 Checking Connection

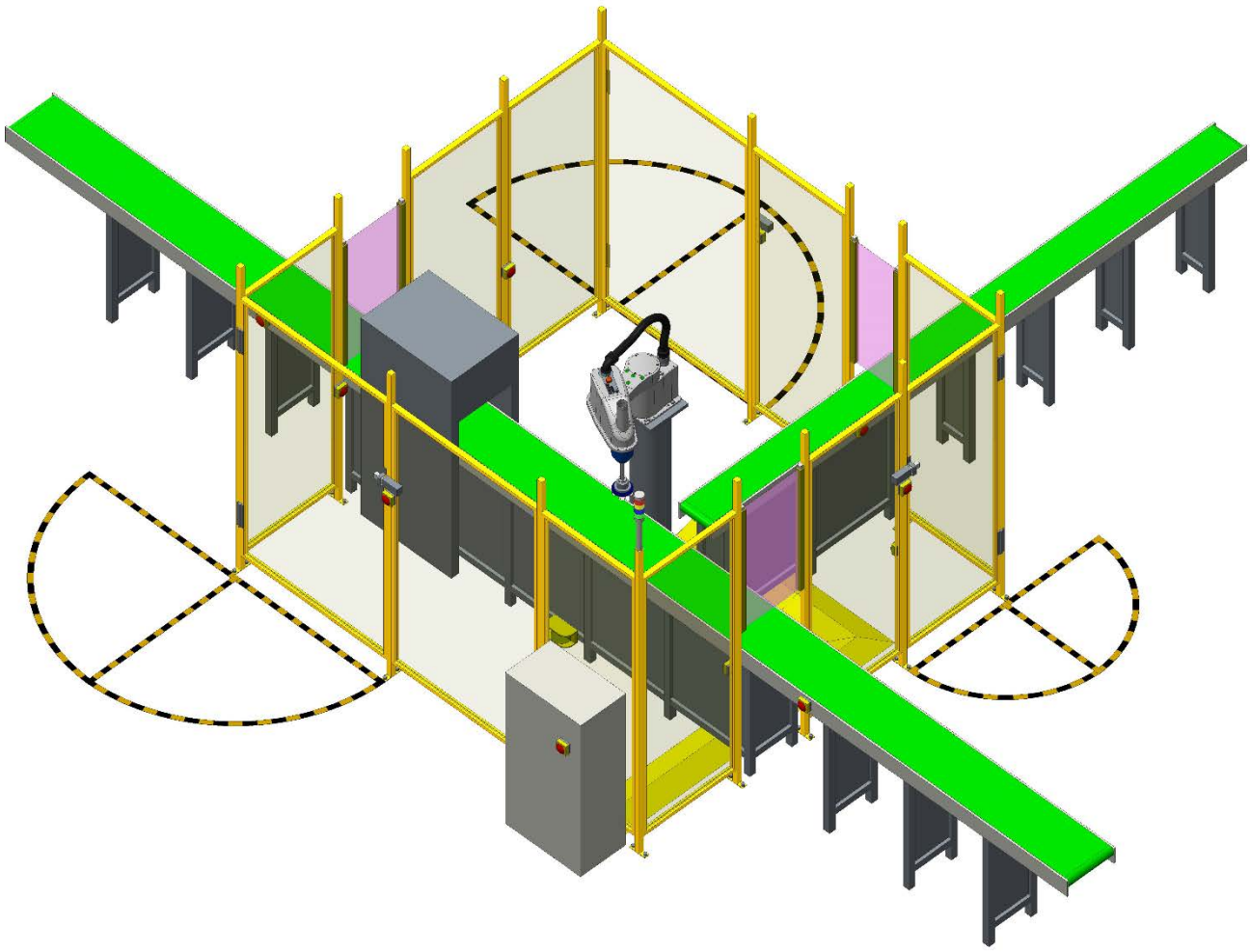
Check the connection with each equipment. Peripheral equipment controlled by the TS5000-EMS controller can operate output signals (on/off) and check input signals using the TP5000 teaching pendant. When operating and checking signals, please refer to the following manual.

- STEA1162 TP5000 Operation Manual, 11.3 I/O Window

It is convenient to register signals that are frequently operated or when operating the solenoid valve with double solenoid specification in the auxiliary signal window.

When registering and operating, please refer to the following manual.

- STEA1162 TP5000 Operation Manual, 11.16 AUX Window



1.5 Teaching

Create a file, teaching and program, and create an "executable program" to operate the robot.

1.5.1 Creating a File

Create a file to be used for program execution.

When creating a file, please refer to the following manual.

- STEA1162 TP5000 Operation Manual, 4.1 Displaying the File Edit Window
  4.4 About Saving/Editing a File

1.5.2 Guiding the Robot

Move the robot.

1.5.2.1 Moving the Robot by Pushing it by Hand

Push the robot with the servo off by hand to move it.

When moving the robot, please refer to the following manual.

- STEA1213 THE400/TS5000 Series Robot Manual, 9.1 Guiding the Robot by Pushing It by Hand

1.5.2.2 Guiding with the Teaching Pendant

Guide the robot with the servo turned on using the teaching pendant. Be sure to work outside the safety protection space when operation accompanies servo on.

When guiding the robot, please refer to the following manual.

- STEA1162 TP5000 Operation Manual, 3.1 Manual Guidance

1.5.3 Teaching

Enter the position data.

When displaying the edit window, please refer to the following manual and then perform operation.

- STEA1162 TP5000 Operation Manual, 4.3.1 Displaying the POINT (Edit) window

1.5.3.1 Entering the Current Position of the Robot

After guiding the robot, enter the current position as POINT type data.

When entering the current position, please refer to the following manual.

- STEA1162 TP5000 Operation Manual, 4.3.3 Teaching the Current Position

1.5.3.2 Entering Position Data such as Dimensions

If you know the dimensions of the position where you want to move the robot, enter such numerical values as POINT type data.

When entering the current position, please refer to the following manual.

- STEA1162 TP5000 Operation Manual, 4.3.2 Editing POINT Type Data

1.5.3.3 Moving to the Entered Position

The teaching pendant guides the robot that has been servo-on to the position of the entered data. Also use it to check if the position data has been entered correctly. Be sure to work outside the safety protection space when operation accompanies servo on.

When guiding the robot, please refer to the following manual.

- STEA1162 TP5000 Operation Manual, 4.3.4 Moving the Teaching Point
  4.3.5 Moving the Teaching Point to the Offset Position
  4.3.6 Moving the Teaching Point in Bypass Operation

1.5.3.4 Entering Load Condition Data

The robot has a function to automatically switch the maximum speed, acceleration/deceleration, and servo gain according to the load conditions. If this function is not used, the vibration when the robot is stopped will increase and the robot life will be shortened.

To enable the function, you need to use the PAYLOAD instruction in the program and enter load type data.

When entering load type data, please refer to the following manual.

- STEA1162 TP5000 Operation Manual, 4.3.12 Displaying the PAYLOAD (Edit) window

1.5.4 Programming

Create a robot operation program.

When creating a program, please refer to the following manuals.

- STEA1162 TP5000 Operation Manual, 4.2 Editing of Global and Program Sections
- TS5000-EMS Robot Language Manual (SCLO2)

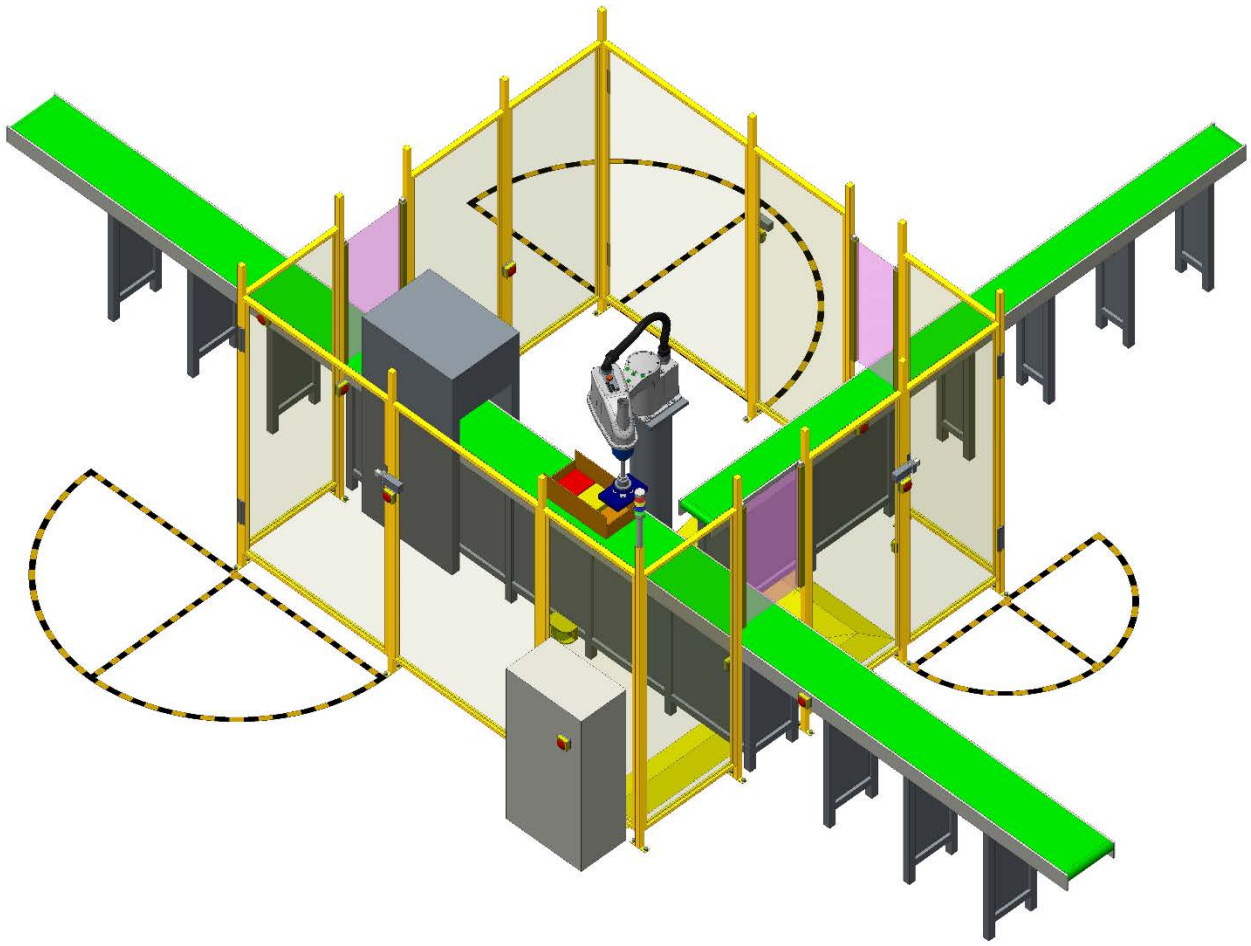
Program examples are provided, so please refer to the following manual.

- TS5000-EMS Robot Language Manual (SCLO2), 4. Program Examples

1.5.5 Test run

Move the robot according to the execution program. In the test run, the robot operates at low speed. When performing a test run, please refer to the following manuals. Be sure to work outside the safety protection space.

- STEA1162 TP5000 Operation Manual, 3.2 About Test Run
- STEA1213 THE400/TS5000 Series Robot Manual, 10.1 Rescue Procedure When Restrained by the Robot

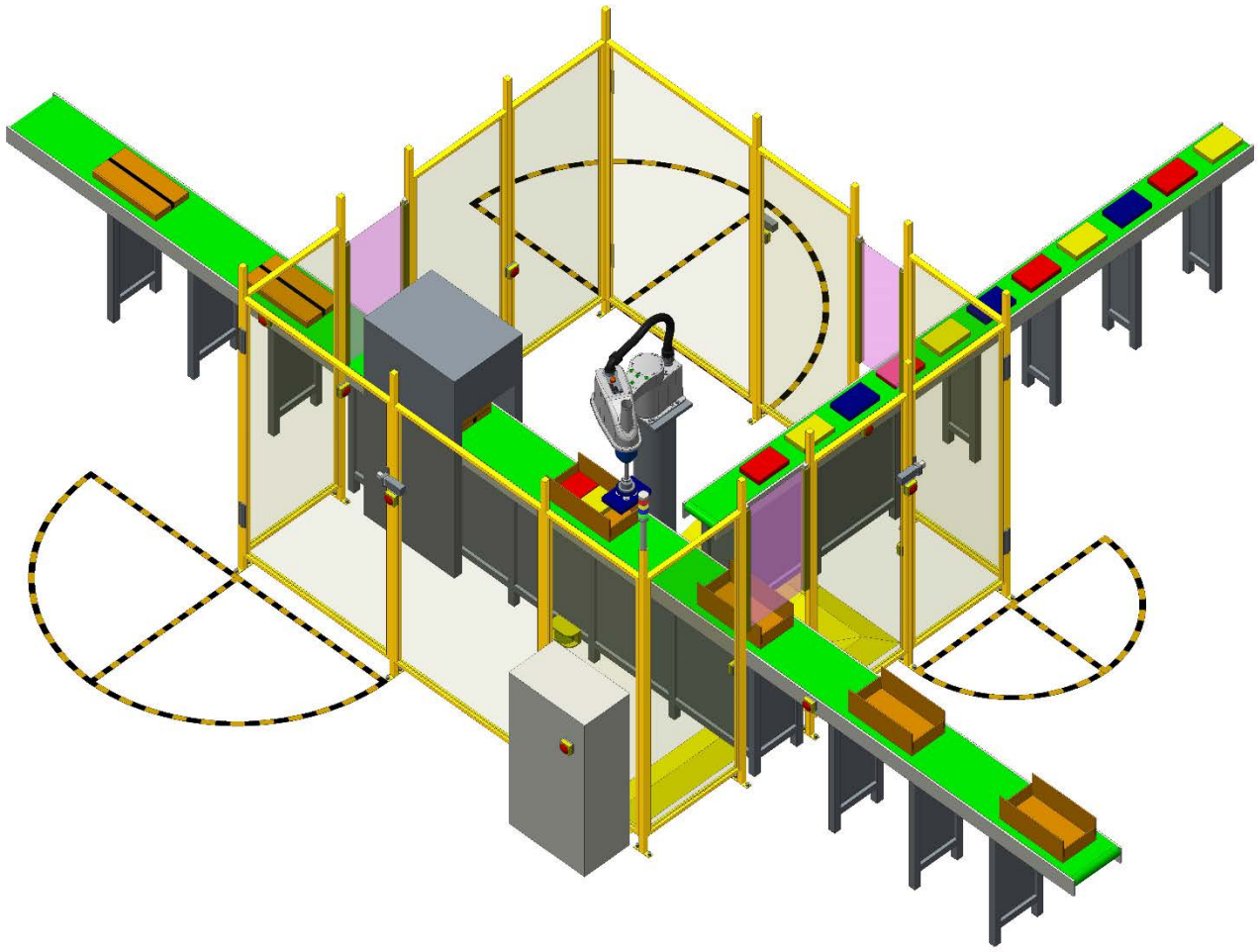


1.6 Operation

Move the robot according to the execution program. Be sure to work outside the safety protection space.

When performing operation, please refer to the following manual.

- STEA1162 TP5000 Operation Manual, 5. Operation in AUTO Mode

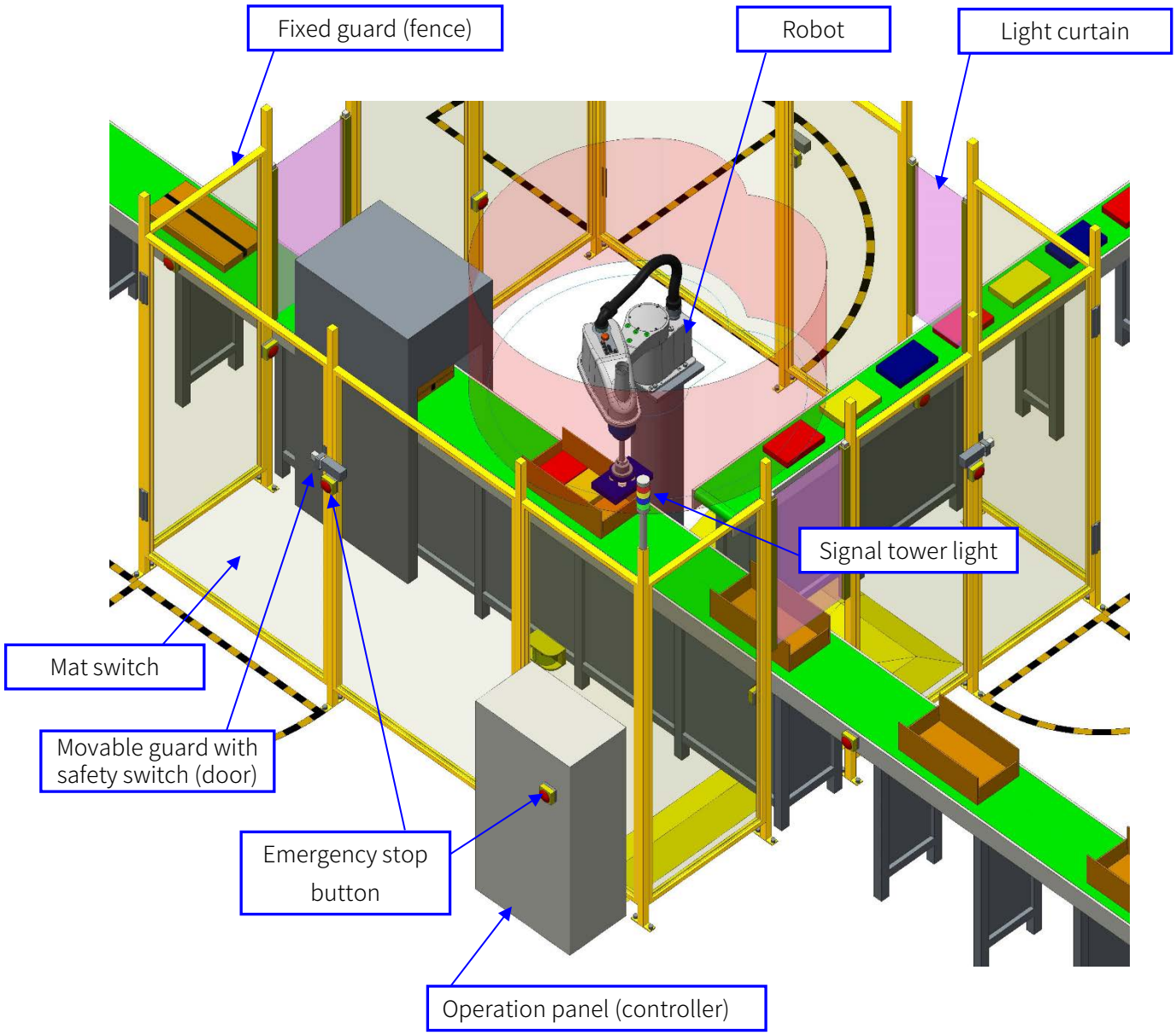


2. Related Standards

Robots and peripheral equipment that make up automated equipment together with robots have standards established for safe use.

Typical standards are listed. Please build safe automation equipment that complies with the applicable standards.

2.1 Typical Related Standards



2.1.1 Robot System Risk Assessment

In the layout design of automated equipment (hereinafter referred to as robot system) that integrates a robot, it is necessary to eliminate hazards and reduce risks by risk assessment.
The standard for risk assessment is as follows.
- ISO 12100 Safety of machinery - General principles for design - Risk assessment and risk reduction

The layout design of the robot system should comply with the following standard.
- ISO 10218-2 Robots and robotic devices - Safety requirements for industrial robots - Part 2: Robot systems and integration

Consider the behavior of the robot during an emergency stop when designing the layout. For detailed performance, please check the following manual.
- STEA1214 THE400-E/TS5000-E Safety Manual, 2.2.2 Robot behavior during emergency stop

2.1.2 Robot

The applicable robot standard is as follows.
- IISO 10218-1 Robots and robotic devices - Safety requirements for industrial robots - Part 1: Robots

This robot complies with ISO10218-1. Some items may not be applicable, so please check the following manuals for detailed specifications and performance.
- STEA1213 THE400/TS5000 Series Robot Manual
- STEA1214 THE400-E/TS5000-E Safety Manual
- STEA1223 TS5000-EMS Controller Manual
- STEA1225 TS5000-EMS Interface Manual
- STEA1162 TP5000 Operation Manual

2.1.3 Guards

By enclosing the movable range of the robot with fixed guards (fences, etc.) and movable guards (doors, etc.), prevent people from easily entering the safety protection space.
The safety performance required for fixed guards and movable guards, and the installation layout standards for avoiding hazard sources are as follows.
- ISO 14120: General requirements for the design and construction of fixed and movable guards
- ISO 14119: Interlocking devices associated with guards
- ISO 13854: Minimum gaps to avoid crushing of parts of the human body
- ISO 13857: Safety distances to prevent hazard zones being reached by upper and lower limbs

2.1.4 Emergency Stop Device

Install the emergency stop device in a place where the operator who has judged the abnormality can immediately push it. The applicable standard is as follows.

- ISO 13850: Emergency Stop

2.1.5 Detection Protection Device

If it is not appropriate to have a guard, you can use a detection protection device such as a light curtain or mat switch. Install a detection protection device so that the robot will stop when a person enters the safety protection space.

The standards for the safety performance required for light curtains and mat switches, and for the distance from the detection area to the dangerous area required for installing protective devices are as follows.

- IEC 61496-1 , -2: Electro-sensitive protective equipment
- ISO 13856-1: Pressure-sensitive protective devices
- ISO 13855: Positioning of safeguards with respect to the approach speeds of parts of the human body

2.1.6 Warning device

Although a warning device does not directly protect people from hazards, it is expected to indirectly prevent disasters by warning machines that they are in a dangerous state or by suppressing erroneous operations.

Use an indicator light that illuminates the operating status of the robot system, or a buzzer that emits sound before a specific status (for example, when the driving power supply is turned on or the operation is started).

It is also possible to suppress erroneous operation by deciding the colors and meanings of the buttons on the operation panel.

The standard for the colors of indicator lights and operation buttons is as follows.

- IEC60204-1: Safety of machinery - Electrical equipment of machines - Part 1: General requirements

Consider noise for audible warnings. Please check the following manual for the noise generated by the robot and the measurement procedure.

- STEA1213 THE400/TS5000 Series Robot Manual, 1.4 Specification Table